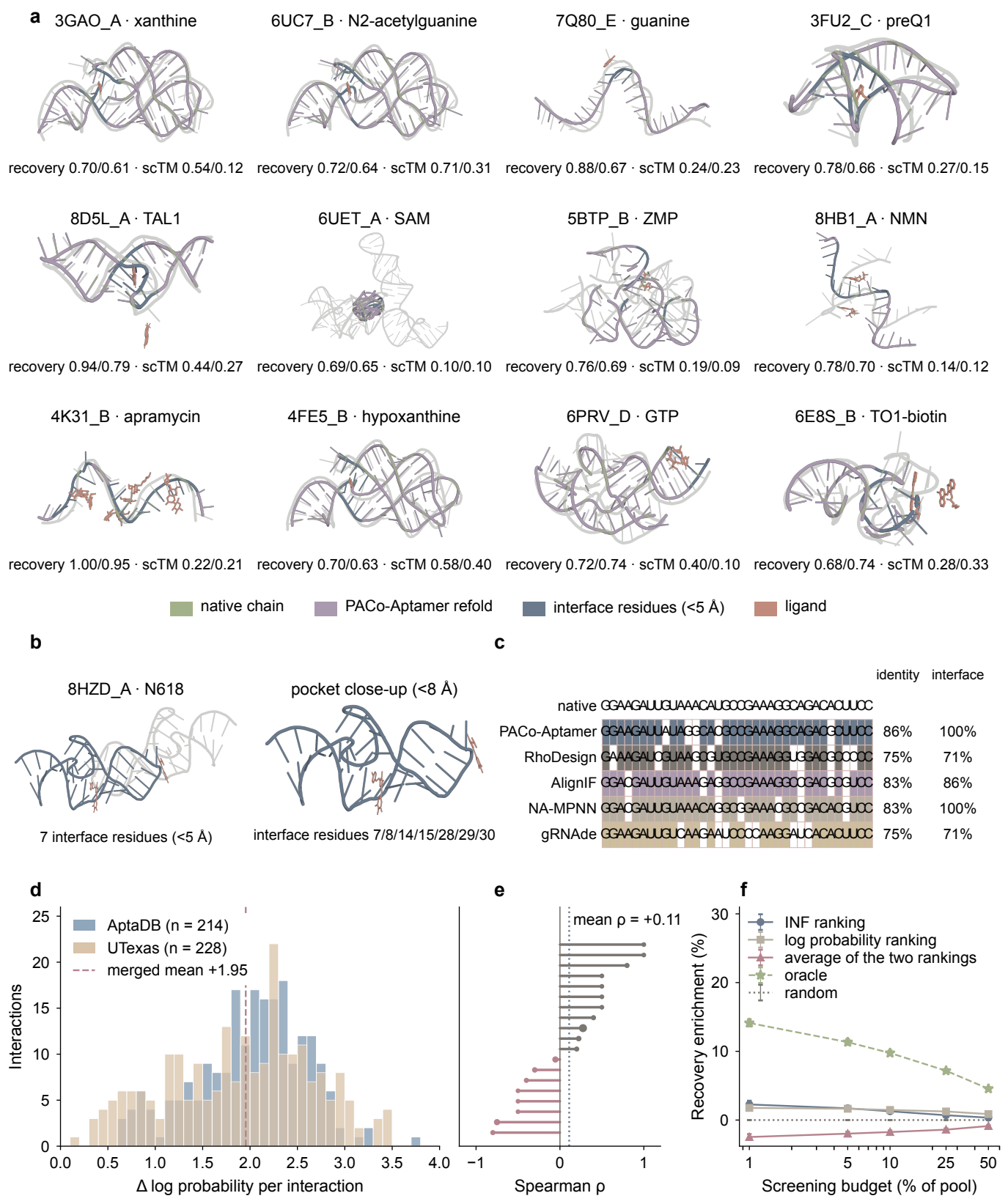

a
3GAO_A · xanthine
6UC7_B · N2-acetylguanine
7Q80_E · guanine
3FU2_C · preQ1
recovery 0.70/0.61 · scTM 0.54/0.12
recovery 0.72/0.64 · scTM 0.71/0.31
recovery 0.88/0.67 · scTM 0.24/0.23
recovery 0.78/0.66 · scTM 0.27/0.15
8D5L_A · TAL1
6UET_A · SAM
5BTP_B · ZMP
8HB1_A · NMN
recovery 0.94/0.79 · scTM 0.44/0.27
recovery 0.69/0.65 · scTM 0.10/0.10
recovery 0.76/0.69 · scTM 0.19/0.09
recovery 0.78/0.70 · scTM 0.14/0.12
4K31_B · apramycin
4FE5_B · hypoxanthine
6PRV_D · GTP
6E8S_B · TO1-biotin
recovery 1.00/0.95 · scTM 0.22/0.21
recovery 0.70/0.63 · scTM 0.58/0.40
recovery 0.72/0.74 · scTM 0.40/0.10
recovery 0.68/0.74 · scTM 0.28/0.33
native chain
PACo-Aptamer refold
interface residues (<5 Å)
ligand
b
8HZD_A · N618
pocket close-up (<8 Å)
7 interface residues (<5 Å)
interface residues 7/8/14/15/28/29/30
c
identity
interface
native GGAAGAUUGUAAACAUGCCGAAAGGCAGACACUUCC
PACo-Aptamer GGAAGAUUAUAGGCAGCCGAAAGGCAGACGCUUCC 86% 100%
RhoDesign GAAAGAUCGUAAGGGUGCCGAAAGGUGGACGCCCCC 75% 71%
AlignIF GGACGAUUGUAAAGAGGCCGAAAGGCCGACGCGUCC 83% 86%
NA-MPNN GGACGAUUGUAAACAGGCGGAAACGCCGACACGUCC 83% 100%
gRNAde GGAAGAUUGUCAAGAAUCCCCAAGGAUCACACUUCC 75% 71%
d
AptaDB (n = 214)
UTexas (n = 228)
merged mean +1.95
Interactions
Δ log probability per interaction
e
mean ρ = +0.11
Spearman ρ
f
INF ranking
log probability ranking
average of the two rankings
oracle
random
Recovery enrichment (%)
Screening budget (% of pool)